

Auxiliary Brake System — Trailer Brake Control (TBC) Module



Trailer Brake Control System

The trailer brake control system consists of the following components:

- Trailer Brake Control (TBC) module, which is mounted in the center of the instrument panel just below the climate controls
- Trailer tow connector
- Instrument Panel Cluster (IPC)

The TBC module has the following driver inputs and outputs:

- Manual slider switch (input)
- Gain buttons (input)
- Gain setting display (output)
- Trailer connectivity icon (output)
- Output bar graph display (output)

The manual slider switch on the TBC module activates the trailer brakes independently from the vehicle brakes. The manual slider is used in conjunction with the gain buttons to adjust and set the trailer brakes. When the manual lever is activated with a trailer connected to the vehicle, the trailer stoplamps and the vehicle stoplamps illuminate. Activating the manual lever without a trailer connected to the vehicle still illuminates the vehicle stoplamps.

The gain buttons set the TBC system for specific towing conditions such as trailer load, vehicle load, road conditions and weather. The gain is normally set to provide maximum trailer braking while maintaining trailer stability. For information on setting the trailer brake gain, refer to the Owner's Literature.

The gain setting and output bar graph displays provide the driver with TBC system information. Trailer connectivity is provided through the trailer connector icon or message center depending on which IPC the vehicle is equipped with, the gain setting display provides the driver with the current gain setting and the output bar graph shows the amount of power going to the trailer brakes relative to brake pedal or manual lever input (only if a trailer is connected to the vehicle).

The TBC module Pulse Width Modulated (PWM) output voltage is speed dependent. With the vehicle stationary and the brake pedal applied, there is approximately 0 volts at the trailer tow connector (pin-3 or pin-7).

TBC System Message Center Messages and Displays

The IPC message center displays the gain setting and relative braking power during a braking event. The message center also indicates trailer connectivity and trailer wiring faults. The following message center messages pertain to the TBC system:

- **NO TRAILER** — This message is displayed whenever the ignition is cycled from OFF to RUN without a trailer connected to the 7-pin trailer connector.
- **TRAILER CONNECTED** — This message is displayed when a correct trailer connection at the 7-pin trailer connector has been made during a given ignition cycle.
- **OUTPUT** — When the vehicle brake pedal is applied, or when the manual control is activated, bar indicators illuminate in the message center to indicate the amount of power going to the trailer brakes relative to the brake pedal or manual control input. One bar indicates the least amount of output with 6 bars indicating maximum output.

- **TRAILER DISCONNECTED** — This message is displayed and accompanied by a single chime when a correct trailer connection was made and then a disconnection has been sensed during a given ignition cycle. This message is also displayed if any of the following occur:
 - the 7-pin trailer connector becomes disconnected while driving
 - an intermittent open circuit occurs in the vehicle wiring from the **TBC** module to the 7-pin trailer connector or the trailer wiring
 - during manual activation using the manual control lever without a trailer connected (accompanied by a single chime)
- **TRAILER BRAKE MODULE FAULT** — This message is displayed and a DTC is stored when a fault is detected by the **TBC** module.
- **WIRING FAULT ON TRAILER** — This message is displayed when the trailer brake output circuit is shorted to ground or shorted to voltage. A DTC sets in the **TBC** module indicating which circuit fault was observed. It is important to determine if a trailer was connected when this message was displayed. If this message is only observed with a trailer connected, the concern is related to the trailer wiring and not the vehicle. If this message is observed without a trailer connected, the concern is with the vehicle wiring from the **TBC** module to the 7-pin trailer connector.
- **TBC FAULT** — This message is displayed if the Instrument Panel Cluster (IPC) is receiving communications from the **TBC** module, but has not been configured for a trailer brake system. If the message center displays **TBC FAULT** and there are no DTCs stored in the **TBC** module, follow the steps in Programmable Parameters to verify and/or modify the **TBC** module programmable parameter in the **IPC**.